WEED RESEARCH An International Journal of Weed Biology, Ecology and Vegetation Management

DOI: 10.1111/wre.12373

Growth traits of the exotic plant *Hydrocotyle vulgaris* and the evenness of resident plant communities are mediated by community age, not species diversity

T-J QIN*, Y-T GUAN*, H QUAN*, B-C DONG*, F-L LUO*, M-X ZHANG*, H-L LI* & F-H YU†

*School of Nature Conservation, Beijing Forestry University, Beijing, China, and †Zhejiang Provincial Key Laboratory of Plant Evolutionary Ecology and Conservation, Taizhou University, Taizhou, China

Received 18 December 2018
Revised version accepted 21 May 2019
Subject Editor: Stephen Novak, Boise State University, Boise, ID, USA

Summary

Biological invasions are determined by interactions between resident plant communities and exotic plants. Time of invasion and species diversity of resident plant communities may greatly affect exotic plant invasions. We assembled low- and high-diversity resident plant communities by sowing seeds of four and eight grassland species, respectively, and at each of three time periods (1, 4 and 7 weeks after sowing), the resident communities were invaded by *Hydrocotyle vulgaris* or not. We also constructed a plant community with *H. vulgaris* alone. Presence of *H. vulgaris* had no effect on biomass of the resident communities or biomass of each component species. Community age significantly affected biomass and evenness of the resident communities, and their competition with *H. vulgaris*, but the priority effect of the resident communities was slight. Increasing species richness did not change the interaction between *H. vulgaris* and the resident plant communities. These findings suggest a weaker competitive exclusionary effect of *H. vulgaris* on the resident communities with early germination, and *H. vulgaris* tended to have no significant impact on intact resident terrestrial plant communities. Thus, the potential risk of *H. vulgaris* invasion is low, especially in the communities with young age.

Keywords: interspecific interactions, invasibility, biological invasions, biomass, competition.

Qin T-J, Guan Y-T, Quan H, Dong B-C, Luo F-L, Zhang M-X, Li H-L & Yu F-H (2019). Growth traits of the exotic plant *Hydrocotyle vulgaris* and the evenness of resident plant communities are mediated by community age, not species diversity. *Weed Research*. https://doi.org/10.1111/wre.12373.

Introduction

An increasing number of invasive alien species are dispersed into and ultimately inhabit resident communities or ecosystems owing to human-related activities (Chaffin *et al*., 2016). Such biological invasions often impair ecosystem structures and functions (Ehrenfeld, 2010; Neve *et al*., 2018). Moreover, the risk of invasions is expected to increase as global climate change intensifies, which may provide new invasion opportunities and reduce biotic resistance of resident communities (Hellmann *et al*., 2008; Diez *et al*., 2012; Hou *et al*., 2014; Sheppard *et al*., 2016). The generalisation of invasion mechanisms remains a challenging task (Ren & Zhang, 2009; Neve *et al*., 2018). There are many mechanisms contributing to biotic resistance, for

Correspondence: H-L Li, School of Nature Conservation, Beijing Forestry University, 35 Qinghua East Road, Haidian District, Beijing 100083, China. Tel: +86-010-62336293; Fax: +86-010-62336724; E-mail: lihongli327@163.com

Tian-Jian Qin and Yu-Ting Guan contributed equally to this work.

© 2019 European Weed Research Society

example the priority effect (i.e. more aged resident plant communities are less prone to invasions) and the species diversity effect (resident plant communities with higher species richness are less prone to invasions; Chytrý *et al.*, 2008; Cleland *et al.*, 2015; Viana *et al.*, 2016).

Some studies found that the early stage of community establishment is more vulnerable to invasions (Symons & Arnott, 2014; Helsen *et al.*, 2016). Moreover, many studies demonstrated that invasive species exert priority effect mainly through their characteristics such as earlier seed germinations, higher survival rates and faster growth rates (Grman & Suding, 2010; Dickson *et al.*, 2012; Wilsey *et al.*, 2015; Goodale & Wilsey, 2018). Therefore, earlier invasive immigrants in a community may significantly affect the establishment of subsequent immigrating species (Grman & Suding, 2010; Helsen *et al.*, 2016), which is important to restore communities and control invasive species. However, priority effect exerted by resident species may also competitively exclude exotic invaders. For example, Cleland *et al.* (2015) observed that resident species outperformed exotic species within a 7-day transplanting period. So far, the general mechanism of the priority effect and the interaction between the priority effect and the species diversity effect opposing plant species invasion remains elusive and uncertain.

The effect of species diversity on the resistance of resident plant communities to exotic plant invasion is also an important topic in ecology (Emery & Gross, 2007; Zhu *et al.*, 2015; Petruzzella *et al.*, 2018). Elton (2000) hypothesised that increasing resident species diversity can decrease invader success. The underlying mechanisms can be attributed to the identity effect and the complementary effect (Clark, 2013). Although numerous experimental and theoretical studies supported this hypothesis (Levine & D'antonio, 1999; Stachowicz *et al.*, 2002; Richardson & Pyšek, 2007; Whitfeld *et al.*, 2014), some other studies came to different results. For example, Altieri *et al.* (2010) and Jauni and Hyvönen (2012) observed positive correlation between species diversity and invader success; Dukes (2001) found that species diversity of grassland communities had no effect on invader's (*Centaurea solstitialis*) success. The relationship between species diversity and invasibility of resident plant community remains controversial.

The priority effect and the species diversity effect may also act simultaneously during community establishment, because the number of species in a given community is expected to increase as new species arrive (Viana *et al.*, 2016). Elucidating mechanisms regulating the strength of priority effects may help further understand variation in diversity–invasibility relationships among natural communities (Goodale & Wilsey, 2018). However, it is unclear how the priority effect and species diversity effect interact to affect the invasibility of resident plant communities.

Here, we conducted a glasshouse experiment to test the effects of species diversity and community age on grassland communities with and without the invasive plant *Hydrocotyle vulgaris*. Specifically, we addressed the following questions. (i) Do community age and species diversity of resident plant communities affect the growth performance of *H. vulgaris*? (ii) Does the effect of invasion by *H. vulgaris* on the growth of resident plant communities decrease with increasing community age and species diversity of the resident communities? (iii) Is the interaction between resident plant communities and *H. vulgaris* affected by community age and species diversity of the resident communities?

Materials and methods

Study species

Hydrocotyle vulgaris L. (Apiaceae), a perennial, clonally reproducing weed, is widely distributed in tropical and temperate regions such as Southeast Asia, Europe and North America. The species was introduced into China as a garden plant in the 1990s and is now considered invasive in China (Miao *et al.*, 2011). It can produce plagiotropic stems, and each node along the stems can form a ramet that consists of one leaf and many adventitious roots. The species can grow in both wet and dry habitats and shows strong phenotypic plasticity (Dong *et al.*, 2014; Wang *et al.*, 2018).

The *H. vulgaris* plants used in this experiment were collected from wetlands in the suburbs of Hangzhou, Zhejiang Province, China, where *H. vulgaris* dominated and then propagated vegetatively in a glasshouse at Forest Science of Beijing Forestry University. Each ramet of *H. vulgaris* we selected for the experiment had one node, one leaf and a few roots. Different populations of *H. vulgaris* were found to have different responses to the environment (Wang *et al.*, 2018). However, we did not assess variation among different populations of *H. vulgaris* in this experimental design, and all ramets used in this study were sampled from a single population.

Resident plant communities

A total of eight species were used to establish resident plant communities, including four grasses (*Setaria viridis*, *Festuca arundinacea*, *Poa pratensis* and *Bromus inermis*), two legumes (*Trifolium repens* and *Astragalus sinicus*) and two forbs (*Plantago asiatica* and *Oxalis*

© 2019 European Weed Research Society

corniculata). All species are common, terrestrial and herbaceous species randomly drawn from a species pool often found in the steppe of northern China (Chen *et al*., 2007; Yang *et al*., 2009; Yin *et al*., 2010; Zhang *et al*., 2010; Miao *et al*., 2011; Yuan *et al*., 2014). These species often co-occur in natural communities (Liu *et al*., 2016). The resident communities were created by sowing seeds of the selected species. Seeds were purchased from Chinese Academy of Agricultural Sciences. Seeds of all species have germination rates higher than 80%.

Experimental design

The experiment had two invasion levels (resident communities with or without invasion by *H. vulgaris*), crossed with three community age levels of resident communities (*H. vulgaris* were transplanted into the communities 1, 4 and 7 weeks after sowing) and two levels of species diversity (resident communities were classified as having low or high species diversity; they were established with four and eight species respectively). Communities with high species diversity were created by sowing all eight species described above, that is four grasses, two legumes and two forbs. Communities with low species diversity were created by randomly drawing two species from the grasses, one species from the legumes, one species from the forbs (thus four species in total) and then sowing the seeds. We sowed all eight species with the same amount of seeds to ensure the same density. We also set up a control treatment for *H. vulgaris* alone with no resident species. There were 13 treatments in total. See Figure S1 for details.

Each treatment was replicated six times and distributed randomly amongst 78 square plastic containers (50 cm long × 50 cm wide × 15 cm deep) filled with 25 L of a 1:1:1 (v/v/v) mixture of quartz sand, vermiculite and peat. On 4 July 2014, 10 August 2014 and 31 August 2014, seeds of each of the four or eight grassland species were sown separately into 7-, 4- and 1-week treatments to simulate different community ages. For the low-diversity treatment, 160 seeds of each of the four species were sown into 12 containers separately at each sowing day. For the high-diversity treatment, 80 seeds of each of the eight species were sown separately into 12 containers at each sowing day. The species combinations of the same diversity level were the same across three community ages for each replicate.

On 7 September 2014, each container in the control treatment and the invaded treatment received nine *H. vulgaris* ramets. Each *H. vulgaris* ramet had one node and one leaf and was about 10 cm high.

Measurements and data analysis

On 7 December 2014, the surviving *H. vulgaris* plants in each container were carefully harvested. We counted number of stem nodes and measured both petiole length and total stem length of *H. vulgaris*. Leaf area was measured using WinFOLIA Pro 2004a (Regent Instruments, Québec, Canada). All *H. vulgaris* plants were separated into leaves, stems and roots, oven-dried at 70°C for 48 h and then weighed. We also harvested all resident species, separated into shoots and roots, similarly dried, and weighed. Total biomass of a resident community is the sum of biomass of all resident plant species in a container.

Pielou's evenness index E was calculated as $E = H/\ln(S)$, where H is the Shannon–Wiener diversity index based on the proportional final dry mass and S is the number of species. H was calculated as $H = -\sum P_i \ln(P_i)$ ($i = 1,2,\ldots S$), where S is the number of resident plant species and P_i is biomass of resident species i divided by biomass of the resident plant community (Whitfeld *et al*., 2014).

The competitive effect of *H. vulgaris* on the resident community was calculated using the following equation: Competitive effect = ln[(biomass of the resident community not invaded by *H. vulgaris*)/(biomass of the resident community invaded by *H. vulgaris*)] (Liu *et al*., 2016). A positive competitive effect suggests strong competition between *H. vulgaris* and the resident plant community, while a negative value indicates that *H. vulgaris* promotes the growth of the community.

Data were analysed using SPSS 19.0 (SPSS Inc., Chicago, IL, USA). We performed two-way ANOVAS to test the effects of community age (7-, 4- and 1-week) and species diversity (low or high) on: (i) the growth performance (total mass, root mass, stem mass, leaf mass, number of nodes, petiole length, total stem length and leaf area) of *H. vulgaris*; (ii) the effects of *H. vulgaris* invasion (presence or absence) and community age on biomass of each resident species in the low- and high-diversity resident communities separately; (iii) the effects of community age and species diversity on the competitive effect of *H. vulgaris* on the resident plant communities. We performed three-way ANOVAS to test the effects of *H. vulgaris* invasion (presence or absence), community age and species diversity on biomass and evenness of the resident plant communities. Before analysis, biomass of *S. viridis*, *F. arundinacea*, *A. sinicus* and *P. asiatica* were transformed into logarithm form to remove heteroscedasticity.

© 2019 European Weed Research Society

Results

Responses of H. vulgaris

The presence of the resident plant communities suppressed the growth performance of *H. vulgaris* (Fig. 1). However, neither community age nor species diversity of the resident plant communities had significant effects on any of the growth traits of *H. vulgaris* (Table S1).

Responses of resident plant communities

Overall, community age had significant effect on both biomass and evenness of the resident communities (Table 1). Biomass of the resident communities increased along increasing community age (from 1 to 7 weeks; Fig. 2A). Evenness was higher in the communities aged 1 week than the communities aged 4 or 7 weeks; evenness of the latter two ages (4 and 7 weeks) did not differ significantly (Fig. 2B). Both *H. vulgaris* invasion and species diversity had no significant effect on biomass or evenness of the resident communities (Table 1).

Table 1 Results of ANOVAS for effects of invasion by *Hydrocotyle vulgaris*, community age and species diversity on biomass and evenness of the resident communities

		Biomass		Evenness	
Effect	d.f.	*F*	*P*	*F*	*P*
Invasion (I)	1, 60	0.08	0.776	0.08	0.772
Age (A)	2, 60	**11.478**	**<0.001**	**11.549**	**<0.001**
Diversity (D)	1, 60	0.53	0.471	0.02	0.877
I × A	2, 60	2.16	0.125	0.76	0.470
I × D	1, 60	1.15	0.289	0.00	0.998
A × D	3, 60	0.33	0.721	0.23	0.795
I × A × D	2, 60	0.52	0.597	0.34	0.711

Values are shown in bold if $P < 0.01$.

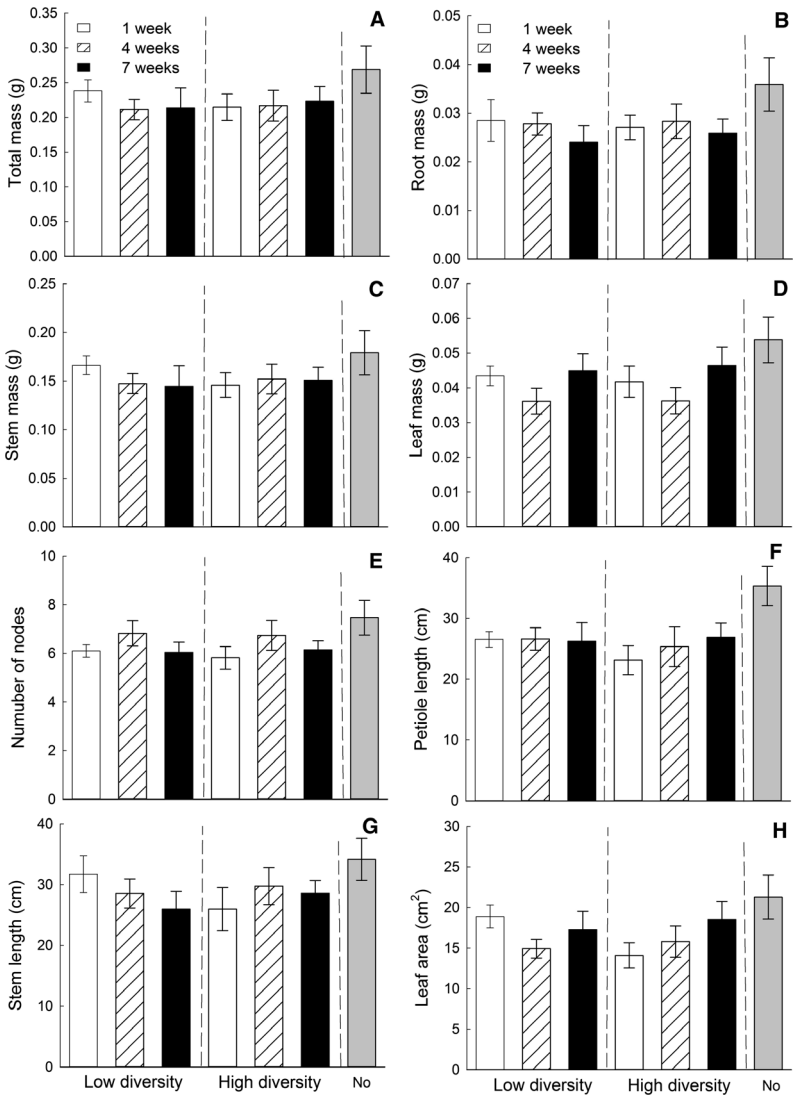

Fig. 1 Effects of community age and species diversity of resident plant communities on the growth of *Hydrocotyle vulgaris* (mean ± SE): (A) total mass; (B) root mass; (C) stem mass; (D) leaf mass; (E) number of nodes; (F) petiole length; (G) stem length; (H) leaf area. See Table S1 for ANOVA summaries. Low diversity and high diversity indicate that *H. vulgaris* and resident plant community of low (four plant species) and high species (eight plant species) diversity were grown in each pot with different community ages respectively. No indicates that there was only *H. vulgaris* in each experimental pot.

© 2019 European Weed Research Society

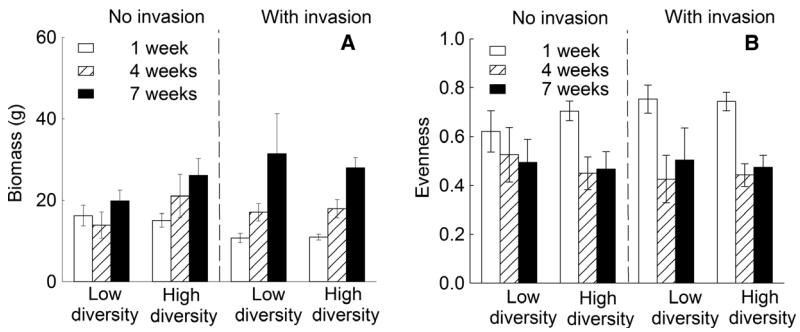

Fig. 2 Effects of community age and species diversity on (A) biomass and (B) evenness of the resident plant communities. Mean ± SE is given. Low diversity and high diversity indicate that *Hydrocotyle vulgaris* and resident plant community of low (four plant species) and high species (eight plant species) diversity were grown in each pot with different community ages respectively.

Responses of each component resident species

Both *H. vulgaris* invasion and its interaction with community age of the resident communities had no significant effects on biomass of any of the component species in either the low- or the high-diversity resident communities (Table 2). Biomass of *S. viridis* in both the low- and high-diversity communities increased with increasing community age of the resident communities. Biomass of *F. arundinacea* in the high-diversity communities increased with increasing community age of the resident communities (Table 2; Figs 3A, B and 4A, B). Biomass of *A. sinicus* and *P. asiatica* in both the low- and high-diversity communities was less in the resident communities aged 4 weeks as compared to the communities aged 1 or 7 weeks. Biomass of *O. corniculata* in the high-diversity communities was less in the resident communities aged 4 weeks as compared to the communities aged 1 or 7 weeks (Table 2; Figs 3F, G and 4F–H). Biomass of *T. repens* in the low-diversity communities was the greatest in the resident communities aged 1 weeks as compared to the communities aged 4 or 7 weeks (Table 2; Fig. 3E).

Competitive effect of H. vulgaris on resident plant communities

Community age of the resident plant communities significantly affected the competitive effect of *H. vulgaris* on the resident communities ($F_{2,30}$ = 4.61; P = 0.018),

Table 2 Results of ANOVAS for effects of invasion by *Hydrocotyle vulgaris* and community age on biomass of each component species in the (A) low- and the (B) high-diversity resident communities

	Invasion (I)		Age (A)		I × A	
Biomass	$F_{1,30}$	P	$F_{2,30}$	P	$F_{1,30}$	P
(A) Low-diversity resident communities						
*Setaria viridis**	0.84	0.378	*5.65*	*0.019*	3.69	0.056
*Festuca arundinacea**	0.54	0.478	2.46	0.127	0.63	0.549
Poa pratensis	0.23	0.643	0.82	0.463	0.12	0.892
Bromus inermis	0.00	0.994	0.37	0.697	0.17	0.848
Trifolium repens	0.29	0.603	**9.00**	**0.004**	1.16	0.347
*Astragalus sinicus**	0.44	0.518	*4.32*	*0.039*	0.16	0.852
*Plantago asiatica**	0.25	0.624	**13.09**	**0.001**	0.35	0.712
Oxalis corniculata	0.03	0.863	2.78	0.102	1.38	0.289
(B) High-diversity resident communities						
*Setaria viridis**	0.37	0.546	**8.70**	**0.001**	0.43	0.653
*Festuca arundinacea**	1.63	0.211	**11.98**	**<0.001**	0.09	0.918
Poa pratensis	0.44	0.511	1.91	0.166	0.39	0.682
Bromus inermis	0.77	0.386	2.18	0.131	0.18	0.841
Trifolium repens	0.43	0.519	1.83	0.178	2.40	0.108
*Astragalus sinicus**	1.45	0.238	**23.24**	**<0.001**	0.07	0.933
*Plantago asiatica**	0.08	0.782	*4.90*	*0.014*	0.74	0.485
Oxalis corniculata	2.09	0.158	**8.02**	**0.002**	1.48	0.245

Values are shown in bold if $P < 0.01$ and in italics if $P < 0.05$.
*log-transformation.

© 2019 European Weed Research Society

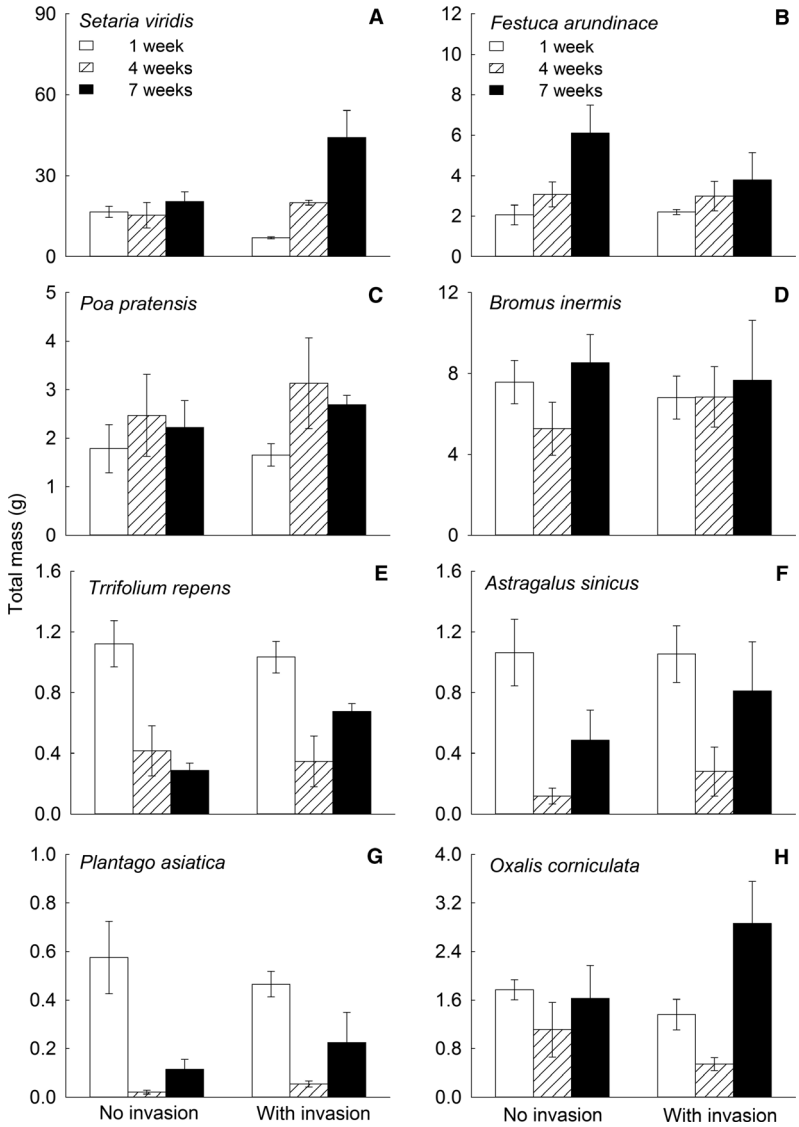

Fig. 3 Effects of invasion by *Hydrocotyle vulgaris* and community age on biomass each component species in the low-diversity resident communities. Mean ± SE is given. No invasion and with invasion indicate that there was only resident plant community, and *H. vulgaris* and resident plant community together in each experimental pot with different community ages (7, 4 and 1 week) respectively.

and with increasing community age the competitive effect changed from positive to negative (Fig. 5). However, neither species diversity ($F_{1,30}$ = 2.24; P = 0.145) nor its interaction with community age ($F_{2,30}$ = 1.20; P = 0.314) had significant effects on the competitive effect (Fig. 5).

Discussion

Effects of community age and species diversity on H. vulgaris growth performance

Some studies have posited that competition and temporal priority can significantly affect establishment success of native grasses (Young *et al.*, 2015; Zuo *et al.*, 2016). In the current study, mean total biomass of the resident species increased along community ages prior to invasion (Table 1; Fig. 2A). These results suggested that resident communities with longer community age may offer less niche space for the invaders and exert stronger competitive effects on the invaders (Wittmann & Fukami, 2018). However, there were no significant differences for all *H. vulgaris* growth traits among the different community age treatments (Fig. 1), suggesting that community resistance did not increase with community age.

Similarly, Wilsey *et al.* (2015) found that resident species only slightly suppressed the establishment of subsequent species, while exotic species, in contrast, had a much stronger priority effect. This finding was also supported by other studies (Dickson *et al.*, 2012; Goodale & Wilsey, 2018). Different invaders may exploit their respective functional niches while invading their respective habitats (Radford, 2013). *Hydrocotyle vulgaris* may occupy functional niches that differ from those of the resident plants; hence, little competition for resources occurred (Fig. 5). Accordingly, *H. vulgaris* traits were not significantly affected by the

© 2019 European Weed Research Society

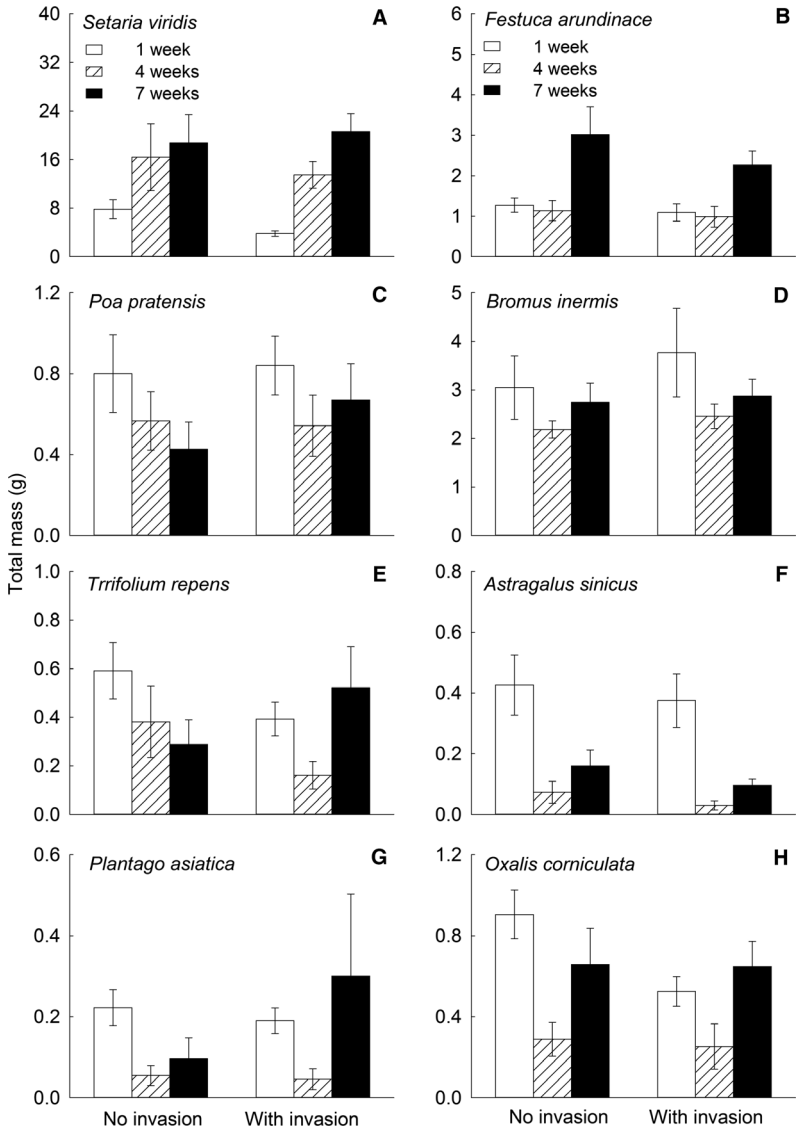

Fig. 4 Effects of invasion by *Hydrocotyle vulgaris* and community age on biomass each component species in the high-diversity resident communities. Mean ± SE is given. No invasion and with invasion indicate that there was only resident plant community, and *H. vulgaris* and resident plant community together in each experimental pot with different community ages (7, 4 and 1 week) respectively.

resident plant community (Fig. 1). The priority effect of the resident communities is slight, and it may suggest that *H. vulgaris* has low potential to invade terrestrial plant communities in northern China.

Effects of presence of H. vulgaris, community age and species diversity on the productivity of resident plant communities and evenness

Our results show that community age had significant effect on both biomass and evenness. The highest biomass was observed in the resident community with 7-week establishment, and the highest evenness occurred in the resident community aged 1 week ($P < 0.01$; Table 1, Fig. 2). These results indicated that interspecific competition among resident species during developmental process caused compositional and structural changes of the community (Cleland *et al.*, 2015; Ali *et al.*, 2018).

At the species level, grass species such as *S. viridis* and *F. arundinacea* tended to accumulate more biomass as time increased (Table 2; Figs 3A, B and 4A, B), while non-grass species produced less in the communities aged 4 weeks (Table 2; Figs 3F–H and 4F–H). The results indicated that competition at different stages had mixed effects on biomass production of the resident species. Grass species, such as *S. viridis*, can better use resources and have higher growth rates (Yang *et al.*, 2009; Jia *et al.*, 2013; Yuan *et al.*, 2014). Thus, their biomass increased faster and competitively became the dominant species as community age increased (Figs 3A, B and 4A, B). The biomass of legumes and forbs was less, and there was no significant difference among community ages (Figs 3 and 4).

© 2019 European Weed Research Society

This may suggest that non-grass species were competitively suppressed by grass species in later community ages. The evenness of resident community aged 4 or 7 weeks became less than that of resident community aged 1 week (Fig. 2B). However, the resident species used in the current study were randomly chosen, and some studies have indicated that different species combinations, established with the same species richness, may produce different results (Altieri *et al.*, 2010; Clark, 2013). In addition, our resident plant communities were established from seeds purchased from the Chinese Academy of Agricultural Sciences, not seeds collected from plants growing in natural communities. The germination rates of naturally occurring seeds may differ from the seeds used in this study, which might change the community structure and evenness values reported here (Cleland *et al.*, 2015).

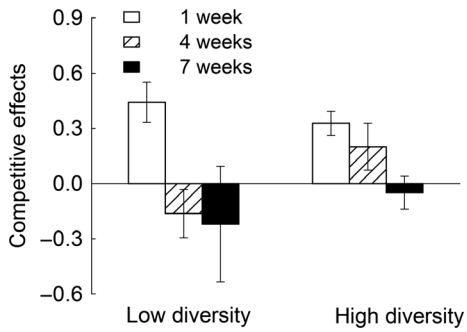

Fig. 5 Effects of community age and species diversity on the competitive effect of *Hydrocotyle vulgaris* on the resident plant communities. Mean ± SE is given. Low diversity and high diversity indicate that *H. vulgaris* and resident plant community of low (four plant species) and high species (eight plant species) diversity were grown in each pot with different community ages respectively.

Effects of community age and species diversity on the interaction between plant community and H. vulgaris

Age of the resident communities had significant effect on the interaction, while species diversity had no significant effect (Fig. 5). This is contradictory to many previous studies (Dukes, 2001; Altieri *et al.*, 2010; Clark, 2013; Wang *et al.*, 2018). Although Elton (2000) postulated that competition in native communities with high species diversity levels should enable such communities to better resist the establishment of invasive species (Richardson & Pyšek, 2007), the results of some studies have suggested that the structure of communities with higher species diversity may be more fragile and therefore more vulnerable to invasions (Stachowicz *et al.*, 2002; Altieri *et al.*, 2010; Jauni & Hyvönen, 2012; Petruzzella *et al.*, 2018). The effect of species diversity may depend on the size of community studied.

Lack of functional diversity, rather than species diversity, possibly played an important role in the current study (Drenovsky *et al.*, 2012). Functional groups have been commonly used to classify plant species based on their morphological and physiological traits. Plant species in different functional groups are usually presumed to acquire resources from different spatial or temporal niches (Davies *et al.*, 2007). Functionally diverse communities may maximise niche complementarity to better resist biological invasions (Pokorny *et al.*, 2005; Frankow-Lindberg, 2012; Clark, 2013). Our study had the same functional groups (grass species, legume species and non-legume forb species) between diversity treatments, and there was no significant difference in total biomasses between two diversity levels (Table 1; Fig. 2A). This suggested that the resource use did not differ between treatments; thus, diversity treatments had no effects on all *H. vulgaris* growth traits and competitive effect. Different plant species prefer different habitat conditions, soil moisture and pot size (Davies *et al.*, 2007). There is evidence that nutrient content can affect *H. vulgaris* growth performance and the interaction between resident community and *H. vulgaris* (Liu *et al.*, 2016). The current study, however, did not take these factors into account, and future studies should investigate these factors.

Conclusions

Community age had significant effect on biomass and evenness of the resident plant communities. The interaction between *H. vulgaris* and resident plant community was mainly mediated by community age, but not species diversity. Maybe functional diversity, more than species richness, plays an important role in the invasibility of a plant community. Additionally, this result suggests that *H. vulgaris* may be less able to invade terrestrial plant communities in northern China. The current study did not take seasonal changes and functional diversity and population difference into account. More studies should be carried out in the future to address the effects of seasonal changes and functional diversity of communities on the invasibility and the competitive interaction between resident plant community and invasive plants such as *H. vulgaris*.

Acknowledgements

We thank Xiang-Qi Bu for his valuable work in the harvest and data processing. We thank Heinz Müller-Schärer and Yan Sun for their valuable comments.

© 2019 European Weed Research Society

This work was funded by National Key Research and Development Program (2016YFC1201100), Technology Program for Water Pollution Control and Treatment (2017ZX07602-004-003), the National Natural Science Foundation of China (31570413) and the Fundamental Research Funds for the Central Universities (2015ZCQ-BH-01).

Conflict of interests

The authors declare no conflict of interests.

Author contributions

H.Q, B.C.D and H.L.L conceived the project and collected data. H.Q, Y.T.G and T.J.Q implemented the experiment; T.J.Q, Y.T.G, B.C.D, H.L.L and F.L.L analysed the data. T.J.Q, Y.T.G, M.X.Z, H.L.L and F.H.Y wrote the first and final drafts. All authors contributed to editing of the manuscript.

References

Ali A, Lohbeck M & Yan E-R (2018) Forest strata-dependent functional evenness explains whole-community aboveground biomass through opposing mechanisms. *Forest Ecology and Management* **424**, 439–447.

Altieri AH, van Wesenbeeck BK, Bertness MD & Silliman BR (2010) Facilitation cascade drives positive relationship between native biodiversity and invasion success. *Ecology* **91**, 1269–1275.

Chaffin BC, Garmestani AS, Angeler DG *et al*. (2016) Biological invasions, ecological resilience and adaptive governance. *Journal of Environmental Management* **183**, 399–407.

Chen M-L, Liu D-Y & Li S (2007) Observation of micromorphological characters of five species in Oxalis. *Journal of Plant Resources and Environment* **16**, 7.

Chytrý M, Maskell LC, Pino J *et al*. (2008) Habitat invasions by alien plants: a quantitative comparison among Mediterranean, subcontinental and oceanic regions of Europe. *Journal of Applied Ecology* **45**, 448–458.

Clark GF (2013) Biodiversity–invasibility mechanisms are mediated by niche dimensionality. *Functional Ecology* **27**, 5–6.

Cleland EE, Esch E & McKinney J (2015) Priority effects vary with species identity and origin in an experiment varying the timing of seed arrival. *Oikos* **124**, 33–40.

Davies KW, Pokorny ML, Sheley RL & James JJ (2007) Influence of plant functional group removal on inorganic soil nitrogen concentrations in native grasslands. *Rangeland Ecology & Management* **60**, 304–310.

Dickson TL, Hopwood JL & Wilsey BJ (2012) Do priority effects benefit invasive plants more than native plants? An experiment with six grassland species *Biological Invasions* **14**, 2617–2624.

Diez JM, D'Antonio CM, Dukes JS *et al*. (2012) Will extreme climatic events facilitate biological invasions? *Frontiers in Ecology and the Environment* **10**, 249–257.

Dong B-C, Wang J-Z, Liu R-H, Zhang M-X, Luo F-L & Yu F-H (2014) Soil heterogeneity affects ramet placement of *Hydrocotyle vulgaris*. *Journal of Plant Ecology* **8**, 91–100.

Drenovsky RE, Grewell BJ, D'antonio CM *et al*. (2012) A functional trait perspective on plant invasion. *Annals of Botany* **110**, 141–153.

Dukes JS (2001) Biodiversity and invasibility in grassland microcosms. *Oecologia* **126**, 563–568.

Ehrenfeld JG (2010) Ecosystem consequences of biological invasions. *Annual Review of Ecology, Evolution, and Systematics* **41**, 59–80.

Elton CS (2000) *The Ecology of Invasions by Animals and Plants*. University of Chicago Press, Chicago, IL.

Emery SM & Gross KL (2007) Dominant species identity, not community evenness, regulates invasion in experimental grassland plant communities. *Ecology* **88**, 954–964.

Frankow-Lindberg BE (2012) Grassland plant species diversity decreases invasion by increasing resource use. *Oecologia* **169**, 793–802.

Goodale KM & Wilsey BJ (2018) Priority effects are affected by precipitation variability and are stronger in exotic than native grassland species. *Plant Ecology* **219**, 429–439.

Grman E & Suding KN (2010) Within-year soil legacies contribute to strong priority effects of exotics on native California grassland communities. *Restoration Ecology* **18**, 664–670.

Hellmann JJ, Byers JE, Bierwagen BG & Dukes JS (2008) Five potential consequences of climate change for invasive species. *Conservation Biology* **22**, 534–543.

Helsen K, Hermy M & Honnay O (2016) A test of priority effect persistence in semi-natural grasslands through the removal of plant functional groups during community assembly. *BMC Ecology* **16**, 22.

Hou Q-Q, Chen B-M, Peng S-L & Chen L-Y (2014) Effects of extreme temperature on seedling establishment of nonnative invasive plants. *Biological Invasions* **16**, 2049–2061.

Jauni M & Hyvönen T (2012) Positive diversity–invasibility relationships across multiple scales in Finnish agricultural habitats. *Biological Invasions* **14**, 1379–1391.

Jia G, Shi S, Wang C *et al*. (2013) Molecular diversity and population structure of Chinese green foxtail [*Setaria viridis* (L.) Beauv.] revealed by microsatellite analysis. *Journal of Experimental Botany* **64**, 3645–3656.

Levine JM & D'antonio CM (1999) Elton revisited: a review of evidence linking diversity and invasibility. *Oikos*, **87**, 15–26.

Liu L, Quan H, Dong B-C *et al*. (2016) Nutrient enrichment alters impacts of *Hydrocotyle vulgaris* invasion on native plant communities. *Scientific Reports* **6**, 39468.

Miao L, Ji M, Wang Y, Qiao D & Chen Y (2011) Study on invasion risk of *Hydrocotyle vulgaris* as an alien species in wetlands (in Chinese). *Journal of Zhejiang University (Agriculture and Life Sciences)* **37**, 425–431.

© 2019 European Weed Research Society

Neve P, Barney J, Buckley Y *et al.* (2018) Reviewing research priorities in weed ecology, evolution and management: a horizon scan. *Weed Research* **58**, 250–258.

Petruzzella A, Manschot J, van Leeuwen CH, Grutters B & Bakker ES (2018) Mechanisms of invasion resistance of aquatic plant communities. *Frontiers in Plant Science* **9**, 134.

Pokorny ML, Sheley RL, Zabinski CA, Engel RE, Svejcar TJ & Borkowski JJ (2005) Plant functional group diversity as a mechanism for invasion resistance. *Restoration Ecology* **13**, 448–459.

Radford IJ (2013) Fluctuating resources, disturbance and plant strategies: diverse mechanisms underlying plant invasions. *Journal of Arid Land* **5**, 284–297.

Ren MX & Zhang QG (2009) The relative generality of plant invasion mechanisms and predicting future invasive plants. *Weed Research* **49**, 449–460.

Richardson DM & Pyšek P (2007) Elton, CS 1958: The ecology of invasions by animals and plants. London: Methuen. *Progress in Physical Geography* **31**, 659–666.

Sheppard CS, Burns BR & Stanley MC (2016) Future-proofing weed management for the effects of climate change: is New Zealand underestimating the risk of increased plant invasions? *New Zealand Journal of Ecology* **40**, 398–405.

Stachowicz JJ, Fried H, Osman RW & Whitlatch RB (2002) Biodiversity, invasion resistance, and marine ecosystem function: reconciling pattern and process. *Ecology* **83**, 2575–2590.

Symons CC & Arnott SE (2014) Timing is everything: priority effects alter community invasibility after disturbance. *Ecology and Evolution* **4**, 397–407.

Viana DS, Cid B, Figuerola J & Santamaría L (2016) Disentangling the roles of diversity resistance and priority effects in community assembly. *Oecologia* **182**, 865–875.

Wang MZ, Bu XQ, Li L, Dong BC, Li HL & Yu FH (2018) Constraints on the evolution of phenotypic plasticity in the clonal plant *Hydrocotyle vulgaris*. *Journal of evolutionary biology* **31**, 1006–1017.

Whitfeld TJ, Roth AM, Lodge AG, Eisenhauer N, Frelich LE & Reich PB (2014) Resident plant diversity and introduced earthworms have contrasting effects on the success of invasive plants. *Biological Invasions* **16**, 2181–2193.

Wilsey BJ, Barber K & Martin LM (2015) Exotic grassland species have stronger priority effects than natives regardless of whether they are cultivated or wild genotypes. *New Phytologist* **205**, 928–937.

Wittmann MJ & Fukami T (2018) Eco-evolutionary buffering: rapid evolution facilitates regional species coexistence despite local priority effects. *The American Naturalist* **191**, E171–E184.

Yang H, Huang Z, Baskin CC *et al.* (2009) Responses of caryopsis germination, early seedling growth and ramet clonal growth of *Bromus inermis* to soil salinity. *Plant and Soil* **316**, 265–275.

Yin JY, Nie SP, Zhou C, Wan Y & Xie MY (2010) Chemical characteristics and antioxidant activities of polysaccharide purified from the seeds of *Plantago asiatica* L. *Journal of the Science of Food and Agriculture* **90**, 210–217.

Young TP, Zefferman EP, Vaughn KJ & Fick S (2015) Initial success of native grasses is contingent on multiple interactions among exotic grass competition, temporal priority, rainfall and site effects. *AoB Plants* **7**, plu081.

Yuan X, Bao Z, Chen Q & Wang G (2014) Development of SCAR marker related to summer stress tolerance in tall fescue (*Festuca arundinacea*). *Notulae Botanicae Horti Agrobotanici Cluj-Napoca* **42**, 81–87.

Zhang X, Zhang Y-J, Yan R, Han J-G, Fuzeng-Hong F & Cao K (2010) Genetic variation of white clover (*Trifolium repens* L.) collections from China detected by morphological traits, RAPD and SSR. *African Journal of Biotechnology* **9**, 3032–3041.

Zhu DH, Wang P, Zhang WZ, Yuan Y, Li B & Wang J (2015) Sampling and complementarity effects of plant diversity on resource use increases the invasion resistance of communities. *PLoS ONE* **10**, e0141559.

Zuo S, Li H, Ma Y & Callaway RM (2016) Temporal priority effects on competition are not consistent among intermountain grassland species. *Acta Oecologica* **75**, 63–67.

Supporting Information

Additional supporting information may be found online in the Supporting Information section at the end of the article.

Figure S1 Illustration of experimental design.

Table S1 Results of ANOVAs for effects of community age and species diversity of resident plant communities on the growth of *H. vulgaris*.

© 2019 European Weed Research Society